

COMP 424 - Artificial Intelligence

Searching Under Uncertainty

Lecture 9: September 30, 2026

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Readings: R&N 4.3, 4.4

Reminders

- A1 is due tonight at 9pm
 - 24-hour grace period without penalty
 - Please don't treat tomorrow 9pm as the deadline! We could close the submission folder anytime after that.
 - You can submit as many times as you'd like. We'll only grade the last submission!
- No class on Monday, Oct 5 because of the provincial election.

The World Is Uncertain!

Major sources of uncertainty:

- Outcomes of actions may be **non-deterministic**
 - E.g., roll a die in a game
- Agent may not have full information about its state
 - **Observable** – know the state
 - **Partially observable** – some information
 - **Non-observable** – don't know at all!

Examples of partial observability

Card games



Blindfolded game?



Word games

G	R	A	I	L
T	R	A	C	K
C	R	A	M	P
C	R	A	B	S
C	R	A	Z	Y
C	R	A	Z	E

Robots with imperfect sensors



Examples of non-deterministic actions

Rolling dice



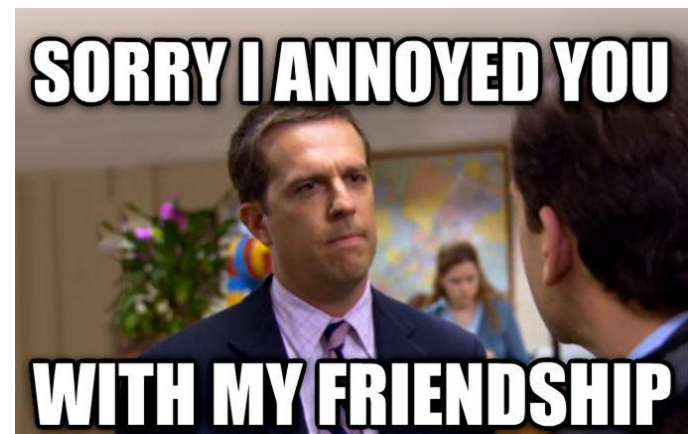
Attacking, etc., in an RPG



Driving on uneven terrain



Anything you say or do as a human being...



Compare To Assumptions In First Lectures

- The environment was **fully observable**
 - The agent always knew exactly what state it's in
- The environment was **deterministic**
 - The outcome of every action is known with certainty
- Consequences:
 - Didn't need to model perceptions
 - Solution to a problem is a single sequence of actions
 - The solution can be predetermined via search
- Module on game-playing broke these assumptions!
So will today's lecture, but in a different way.

Key Questions

- How do we integrate uncertainty into our search process?
 - How do we represent the current progress in our search algorithm?
Can't assume we know what state we're in.
 - Future states cannot be determined in advance!
- What does a solution to the search problem look like?
 - No longer a path.
 - Instead, a **contingency plan** (a.k.a., **policy** or **strategy**)
- We'd still like to guarantee reaching a goal state, if possible!

Today's Sample Problem

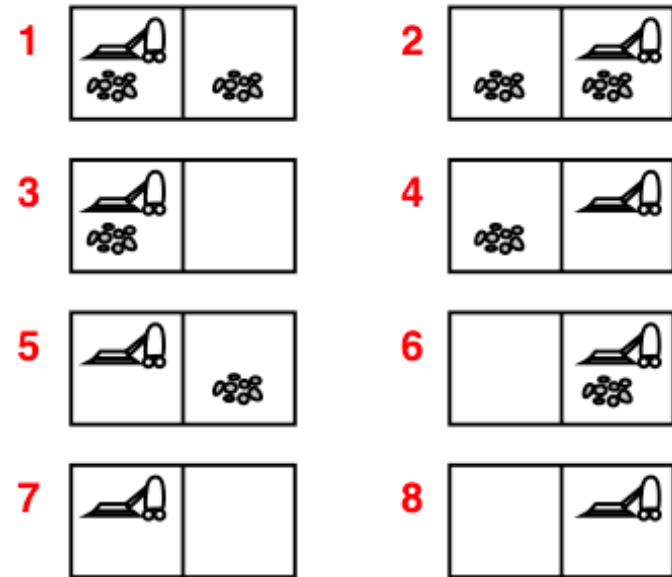


Today's Sample Problem

Let's consider **the vacuum world**:

- **State:**
 - Two rooms can each be clean or dirty
 - Vacuum's location (in one room)
- **Actions:**
 - Move (L/R)
 - Sweep (S)
- **Goal:** to clean both rooms
- **Uncertainty sources:**
 - Vacuum's location (obs/not)
 - Cleanliness status (obs/not)
 - Outcomes of actions are either certain or "noisy"

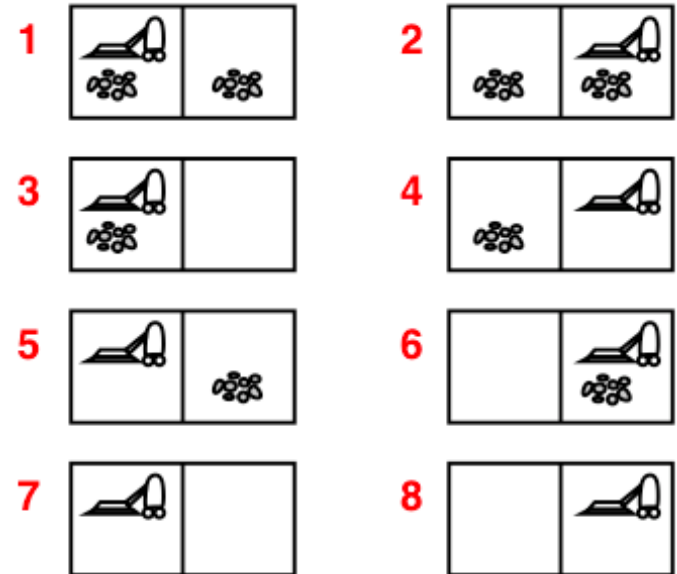
This is the full list
of the 8 possible
underlying states



Observable vacuum world

1. Observable, deterministic case:

- Start in a given initial state: {5}
- Plan: {Right; Sweep}



Actions:

{Left, Right, Sweep}

Transitions: As expected

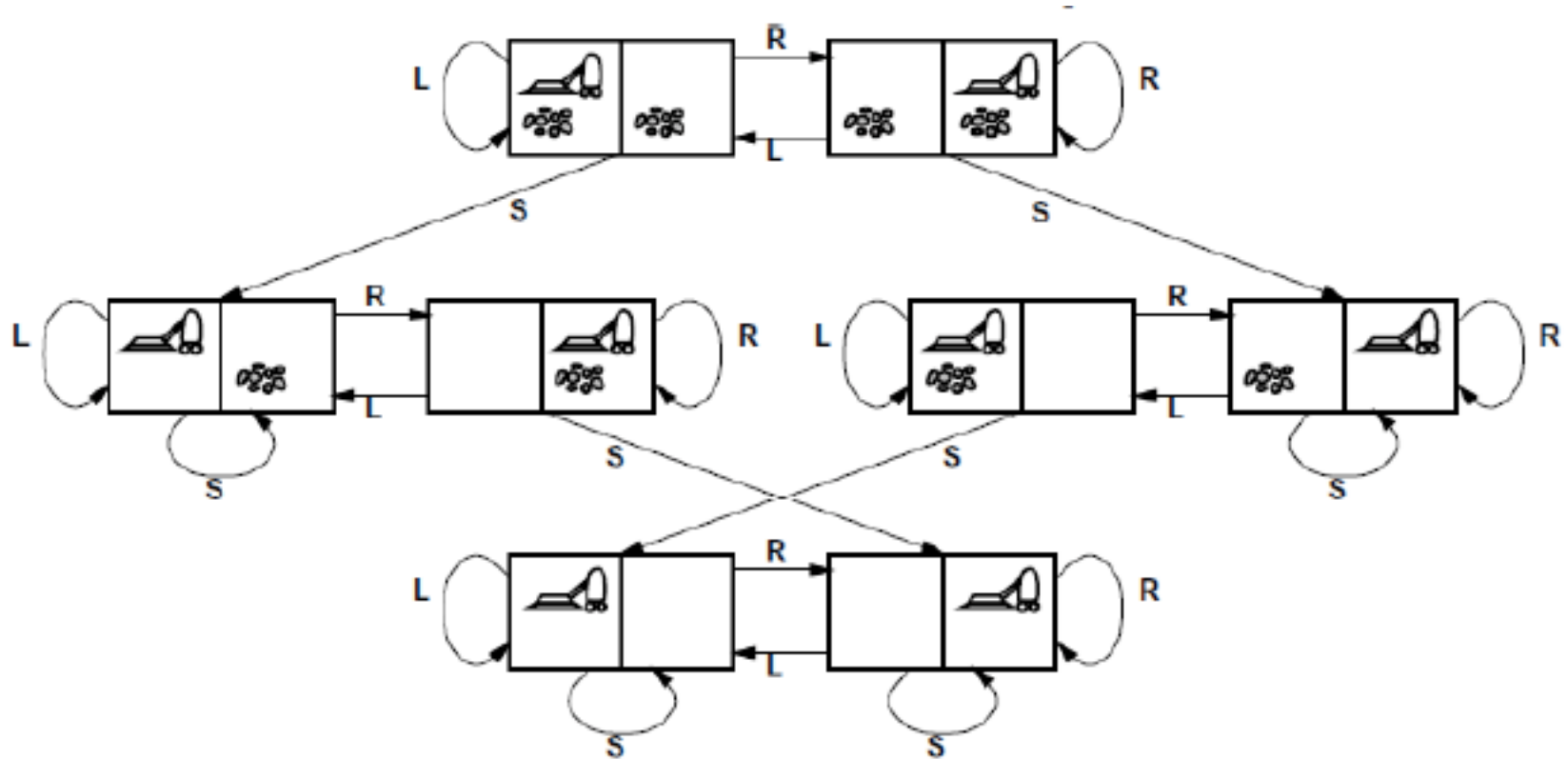
Goal states: {7, 8}

Observable vacuum world: state space

Setting:

Observable

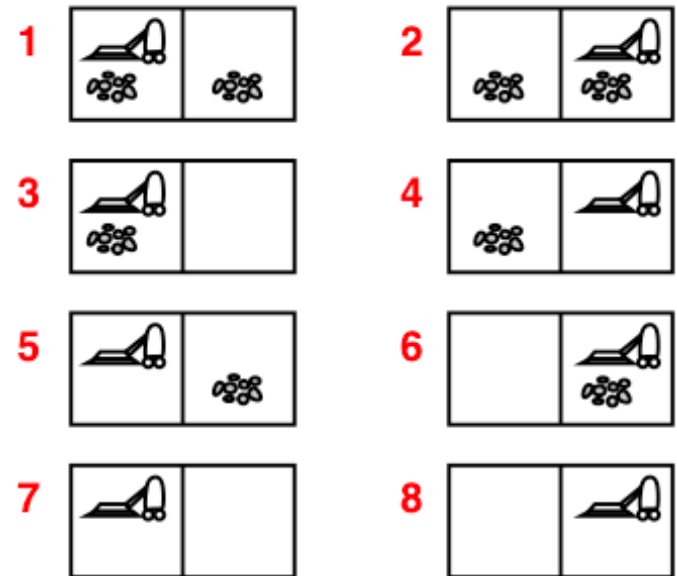
Deterministic actions



Non-observable vacuum world

2. Non-observable case:

- Start in any state: {1,2,3,4,5,6,7,8}
- Agent has **no way to observe state**
 - No sensors, “blind” execution
- Plan: ?



Actions:

{Left, Right, Sweep}

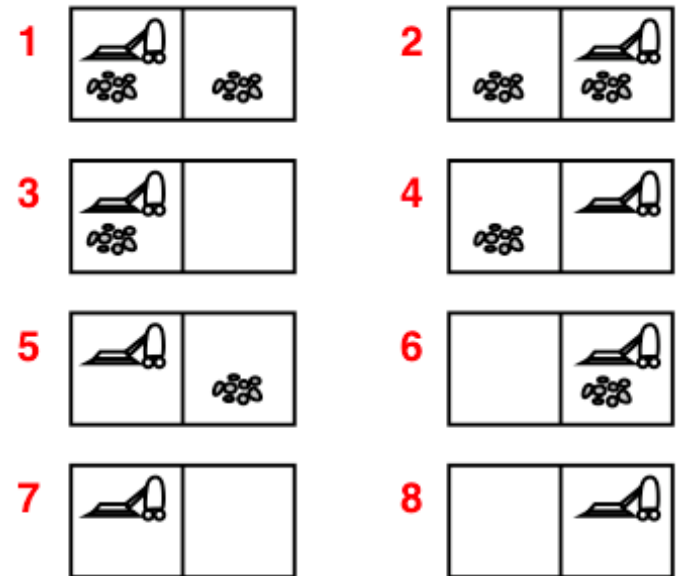
Transitions: As expected

Goal states: {7, 8}

Non-observable vacuum world

2. Non-observable case:

- Start state **set**: {1,2,3,4,5,6,7,8}
- Agent has **no way to observe state**
- Plan: {Right; ? }
- Taking an action (e.g., **Right**) moves the agent to a new **set** of states
- We must reason over **sets** now
- We call these sets “**beliefs**”



Actions:

{Left, Right, Sweep}

Transitions: As expected

Goal states: {7, 8}

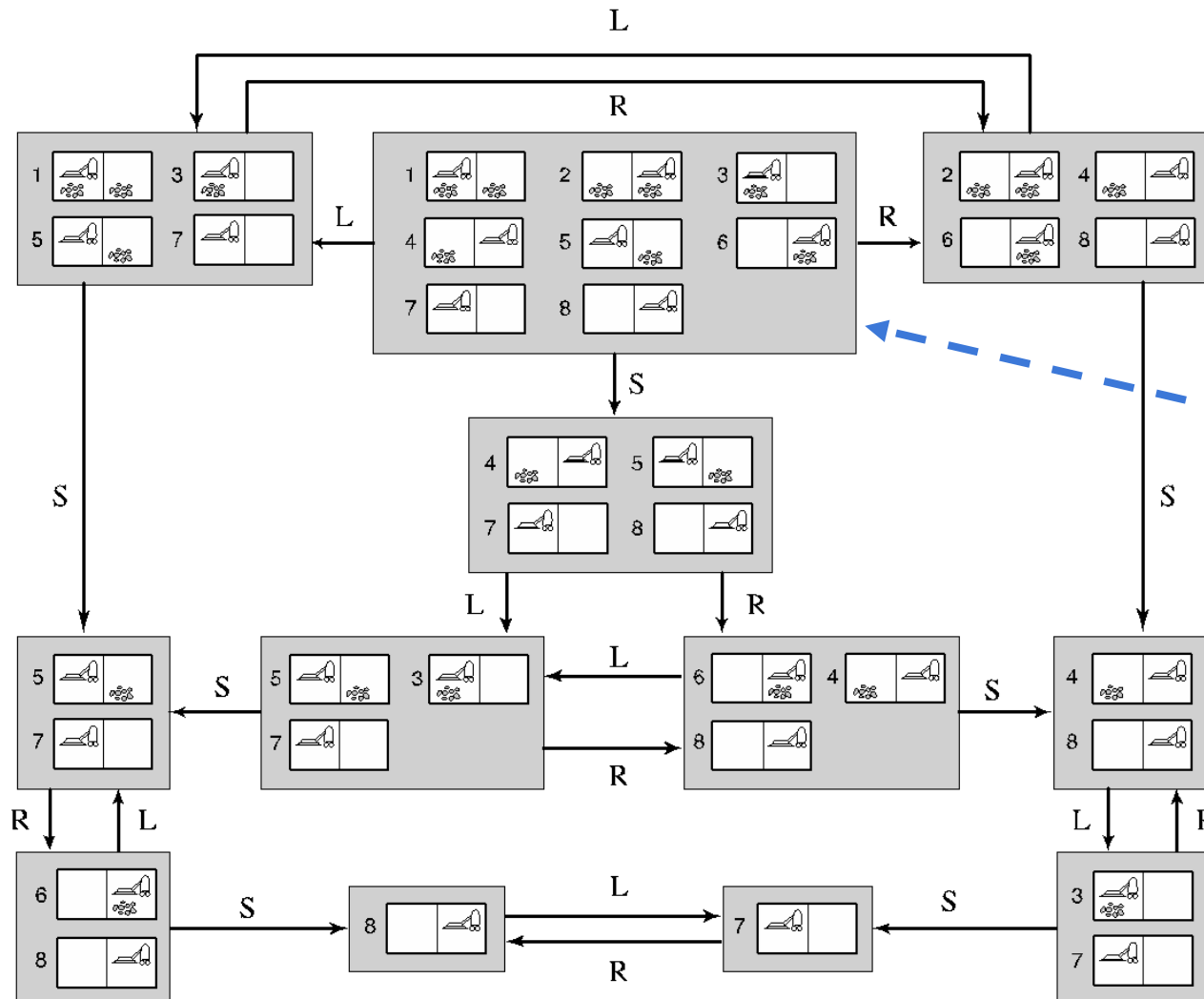
Belief States

- A **belief state** is the set of possible physical states an agent might be in.
 - i.e., the agent's current belief about the set of states that are possible, given the sequence of actions and observations
- What's the total number of distinct belief states?
 - Each of the 8 possible states may be included (or not) in the belief state
 - We can think of a belief state as an 8-bit vector: $2^8 = 256$
 - *Technically 2^8-1 , since the belief has to include at least 1 state*

Searching with unobservable states

- What's the total number of possible beliefs? 2^8-1
 - The representational power of the belief formulation
- But what's the number of **reachable** belief states?
 - Note that there are many fewer underlying states and actions compared to the large number of beliefs.
 - Need to consider the problem dynamics. Some states may be mutually exclusive and other sub-sets will be impossible to reach from the starting belief.

Non-observable vacuum: reachable belief space

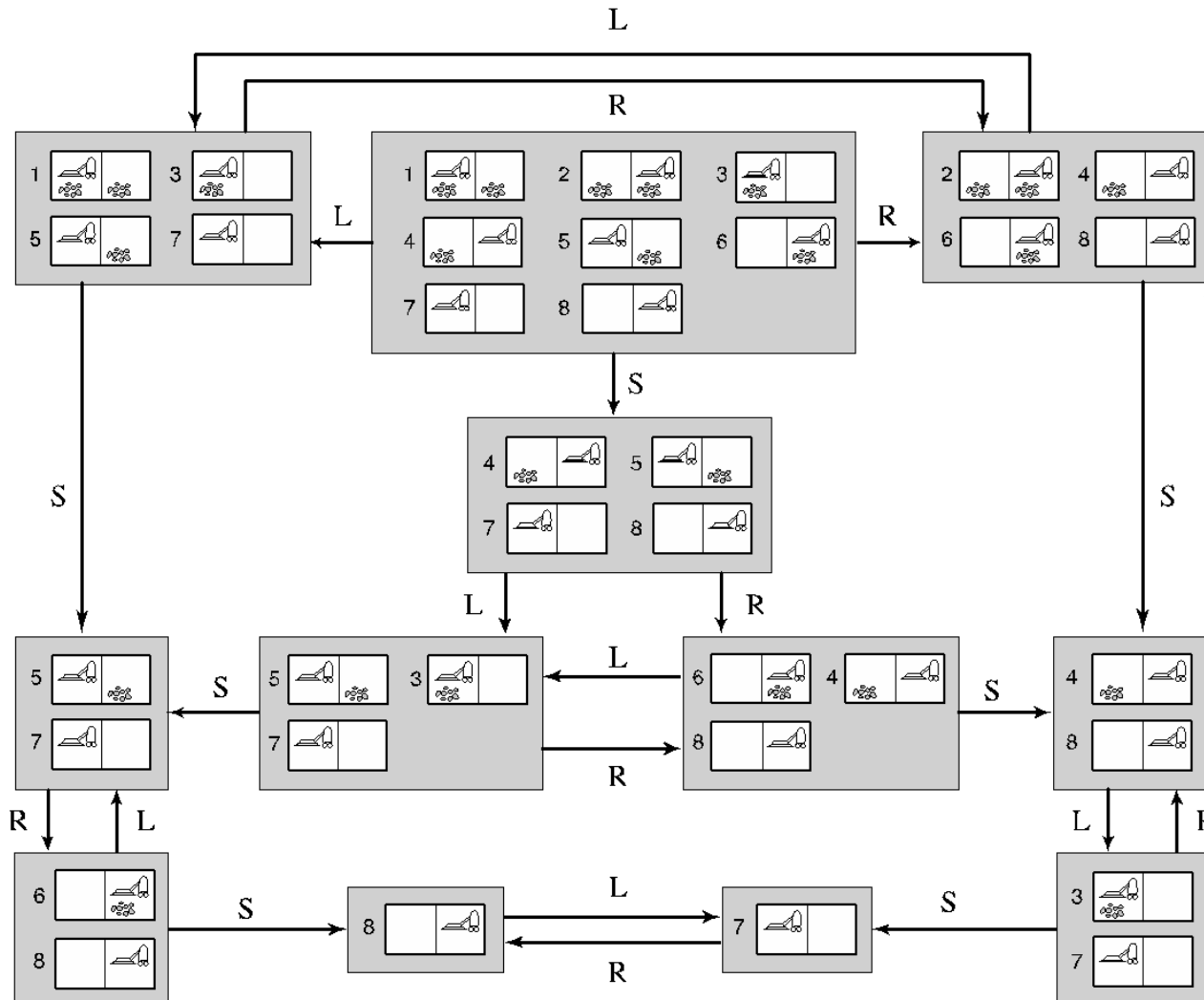


Setting:
Non-observable
Deterministic actions

Initial belief state
=
total ignorance
= 
=
8 physical states

*NB: this figure
omits self links!*

Non-observable vacuum: reachable belief space

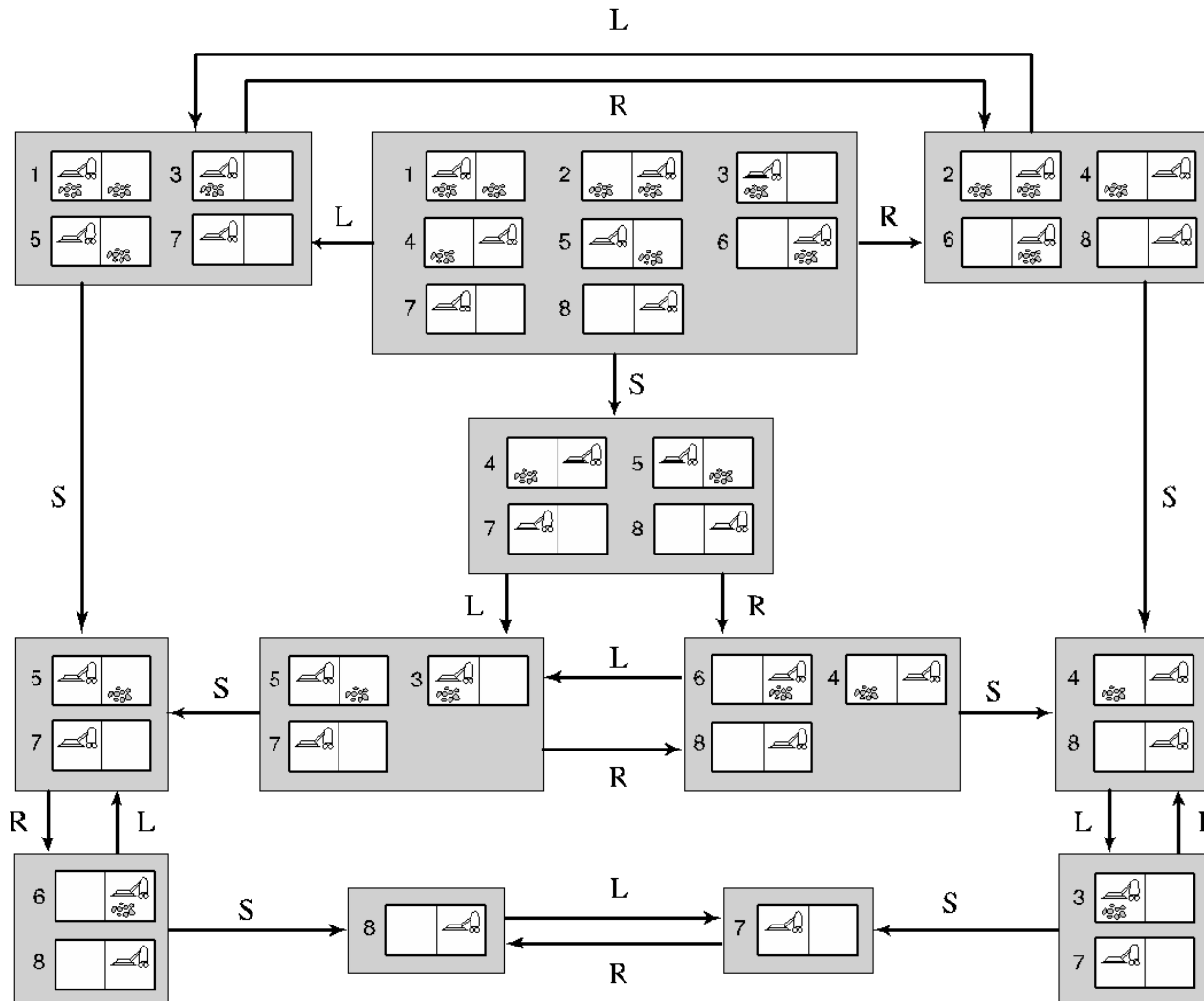


Setting:
Non-observable
Deterministic actions

Result of Right
=
Vacuum is in the
right room
=
4 physical states

*NB: this figure
omits self links!*

Non-observable vacuum: reachable belief space

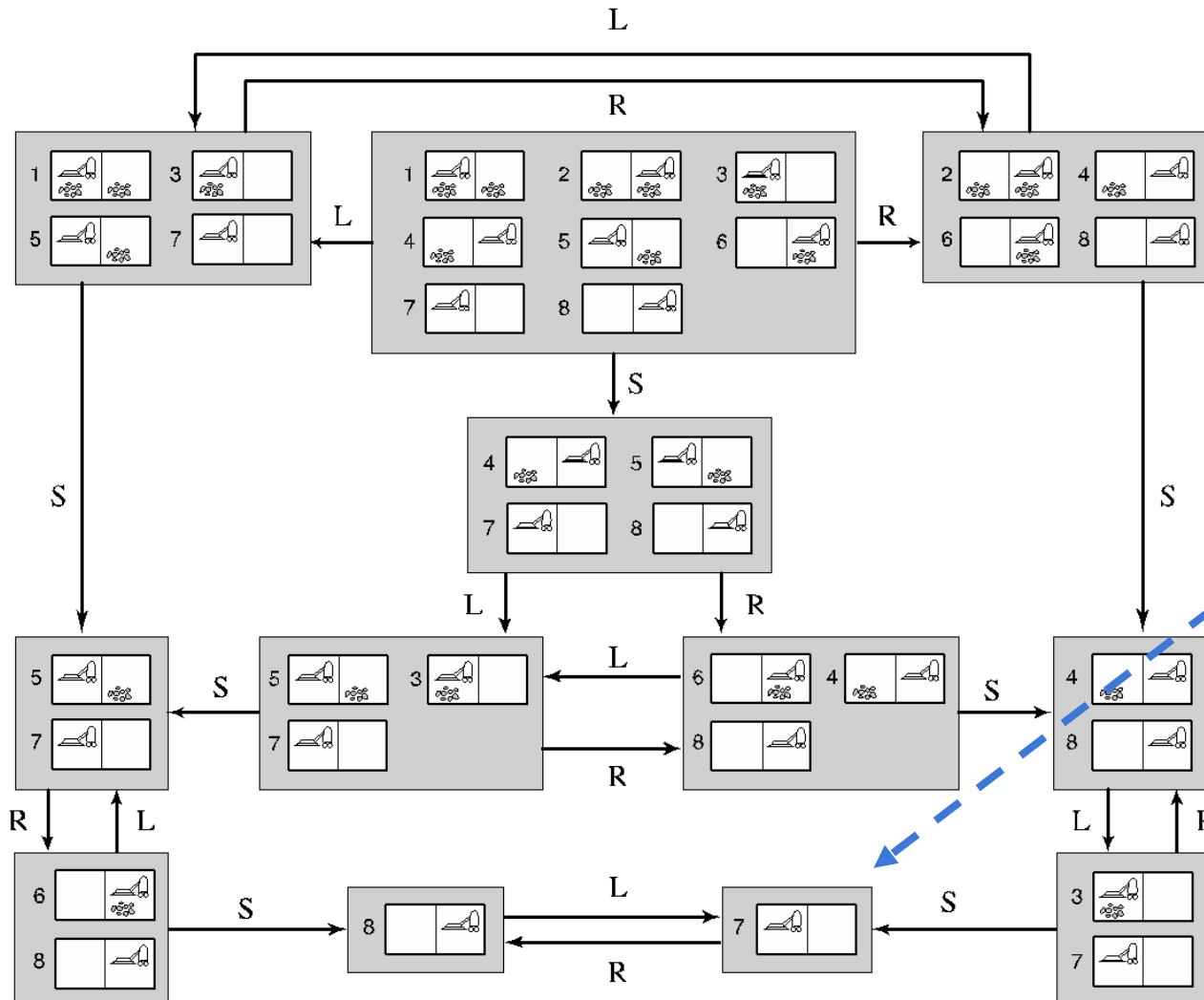


Setting:
Non-observable
Deterministic actions

Continue with a
Sweep
=
We're still in right
room, and it's
clean
=
2 physical states

*NB: this figure
omits self links!*

Non-observable vacuum: reachable belief space



Setting:
Non-observable
Deterministic actions

Full path R,S,L,S
=
We're back to left
room. Both rooms
are now clean
=
1 physical state, a
goal state

*NB: this figure
omits self links!*

Searching with unobservable states

- Recall 2^8-1 belief states by counting the binary membership
- However, the number of **reachable** belief states is **only 12**
 - Planning over this space can be reasonably efficient. Works on the belief-space graph shown previously.
 - Note that these plans are “**open-loop**”. They do not adapt at all to which underlying state we’re in. That’s still unobserved.

Searching with unobservable states

- We have no observations, but actions are (so far) deterministic.

- Start in any state:

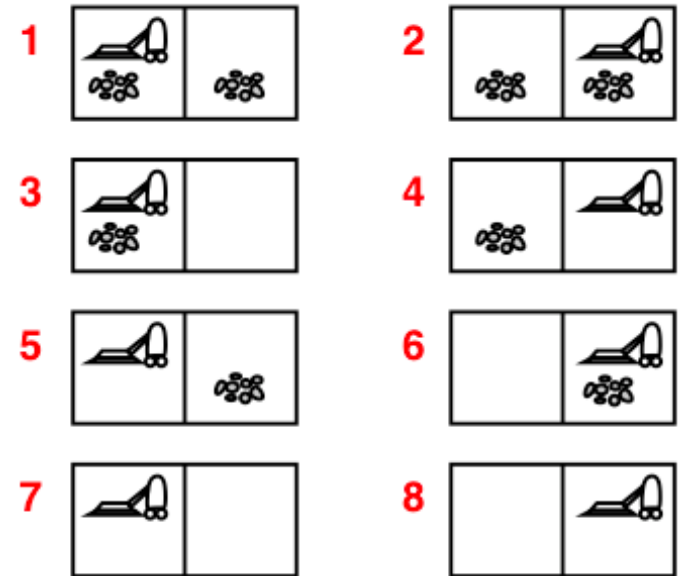
$$b = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

- How to use actions?

$$b = \{1, 2, 3, 4, 5, 6, 7, 8\} \rightarrow_{\text{Right}} \{2, 4, 6, 8\} = b'$$

For deterministic actions: $|b'| \leq |b|$

- Plan?



Searching with unobservable states

- We have no observations, but actions are (so far) deterministic.
- Start in any state:

$$b = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

- How to use actions?

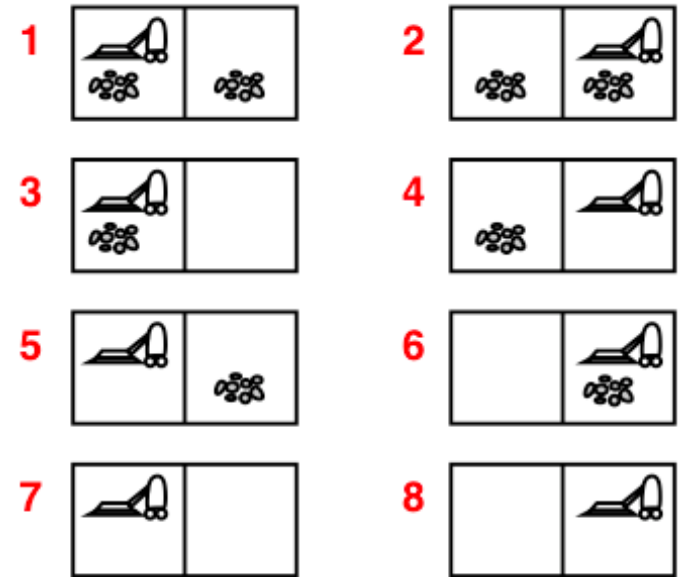
$$b = \{1, 2, 3, 4, 5, 6, 7, 8\} \xrightarrow{\text{Right}} \{2, 4, 6, 8\} = b'$$

For deterministic actions: $|b'| \leq |b|$

- Plan? Use a **conformant plan**

$$\{1, 2, 3, 4, 5, 6, 7, 8\} \xrightarrow{\text{Right}} \{2, 4, 6, 8\} \xrightarrow{\text{Sweep}} \{4, 8\} \xrightarrow{\text{Left}} \{3, 7\} \xrightarrow{\text{Sweep}} \{7\}$$

From a state of total uncertainty, we can **coerce** the world into state **{7}**



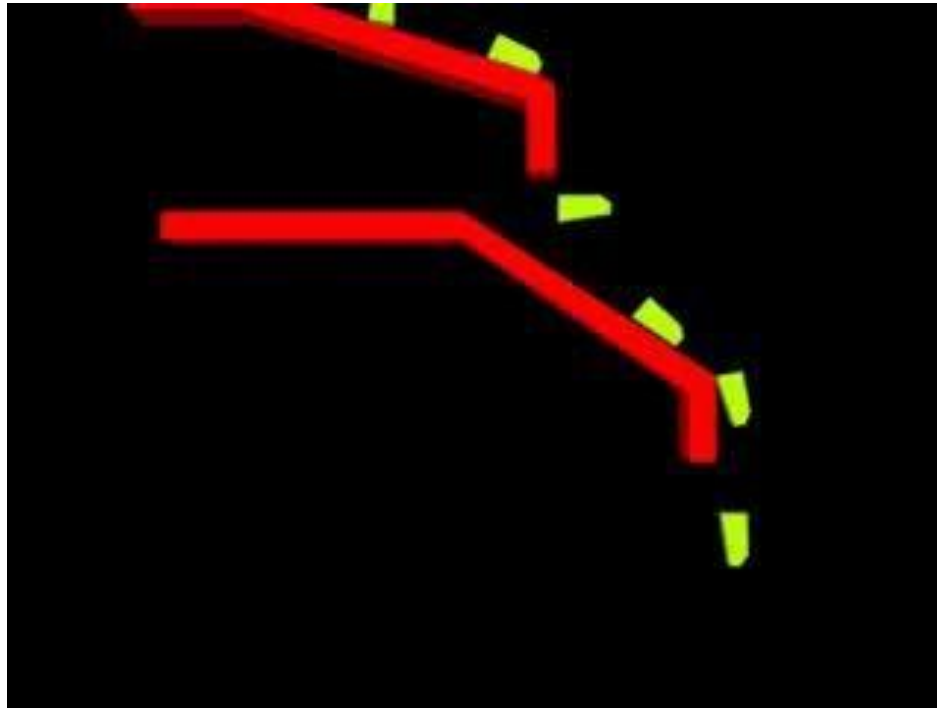
Conformant planning

- **Goal**: a plan that is guaranteed to take us from any initial state to a goal state, no matter what the effect of actions is
- A good search heuristic is to favor actions that **reduce uncertainty**
 - reduce the **belief state** to a single **physical state**
 - you might need to back up from bad belief states
- Once you know what physical state you're in, apply standard search to reach the goal!
- Use **conformant planning** when there are NO observations

Visual analogy

- Feeding parts for manufacturing:

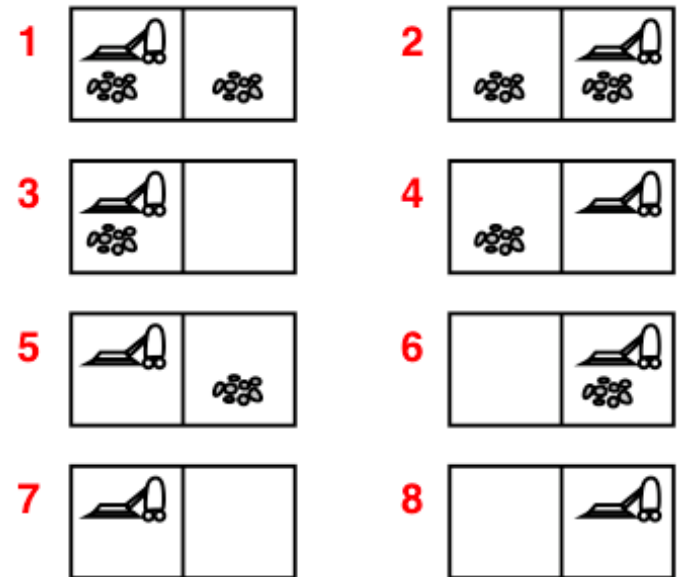
<http://goldberg.berkeley.edu/fences.mpg>



Faulty vacuum world

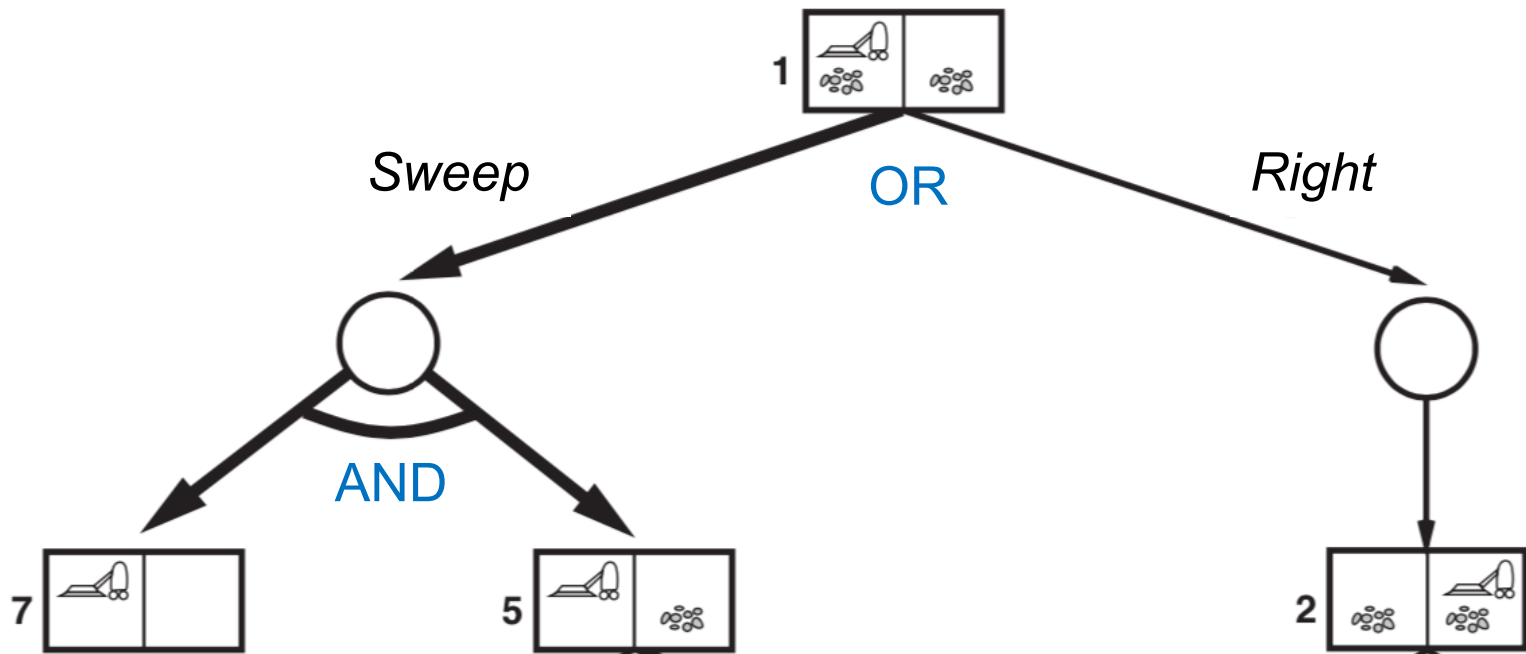
3. Non-deterministic (but observable) case:

- Start in a given state: {1}
- Transitions are no longer as fully predictable:
 - **Sweep**ing in a dirty square sometimes also cleans the adjacent square;
 - **Sweep**ing in a clean square sometimes deposits dirt
- Plan?



Non-deterministic actions: AND-OR search

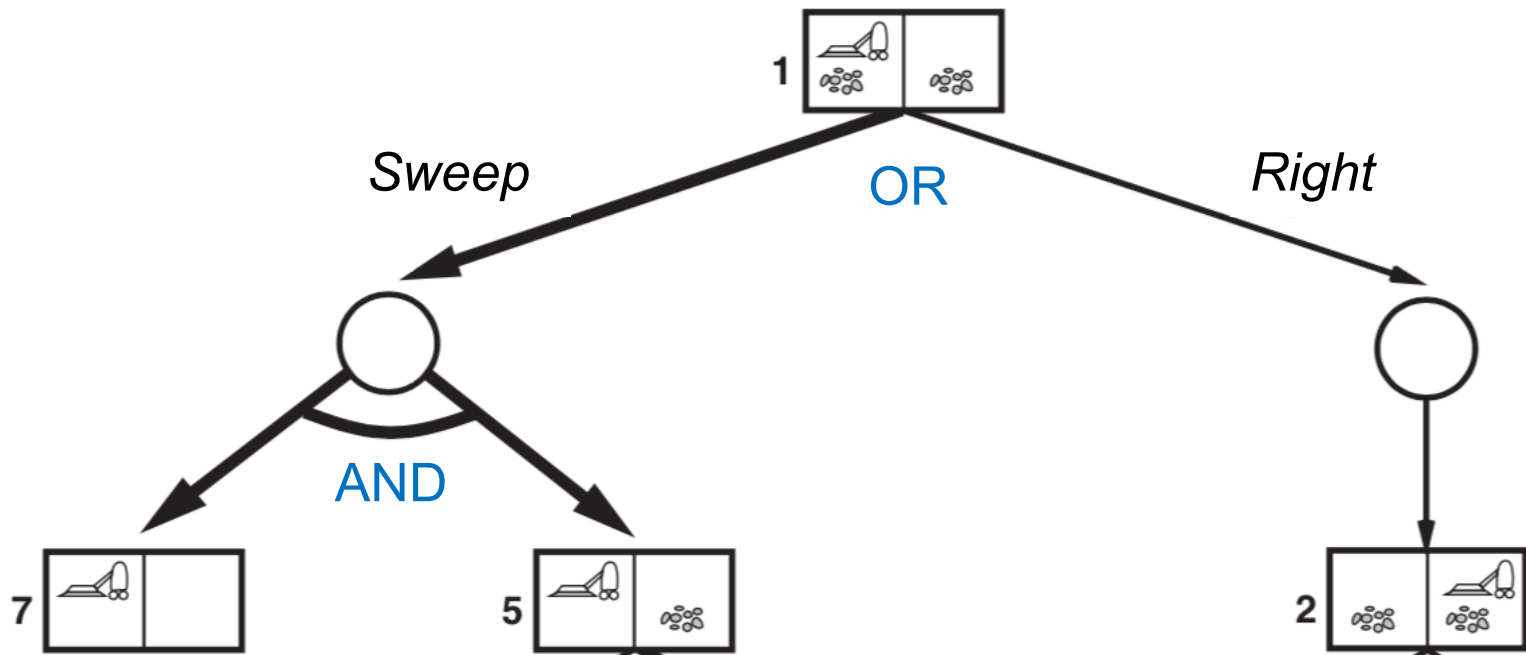
- In the case of non-deterministic actions, we also start by constructing a search tree
- We augment trees to use two node types: **AND** and **OR**

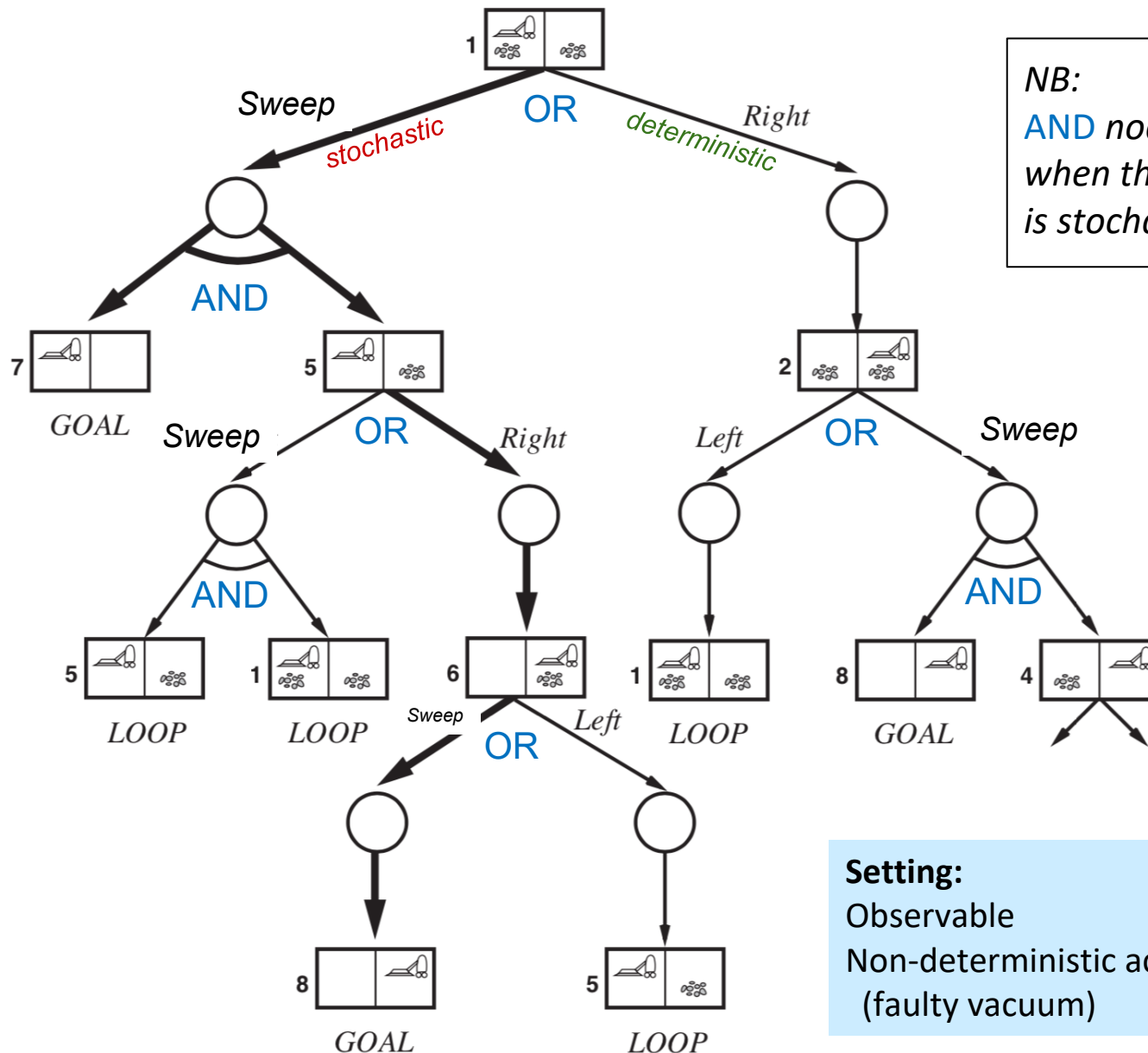


Non-deterministic actions: AND-OR search

AND-OR search tree:

- **OR nodes:** branching is induced by the **agent's** choice between actions
- **AND nodes:** branching is induced by the **environment's** choice of outcome





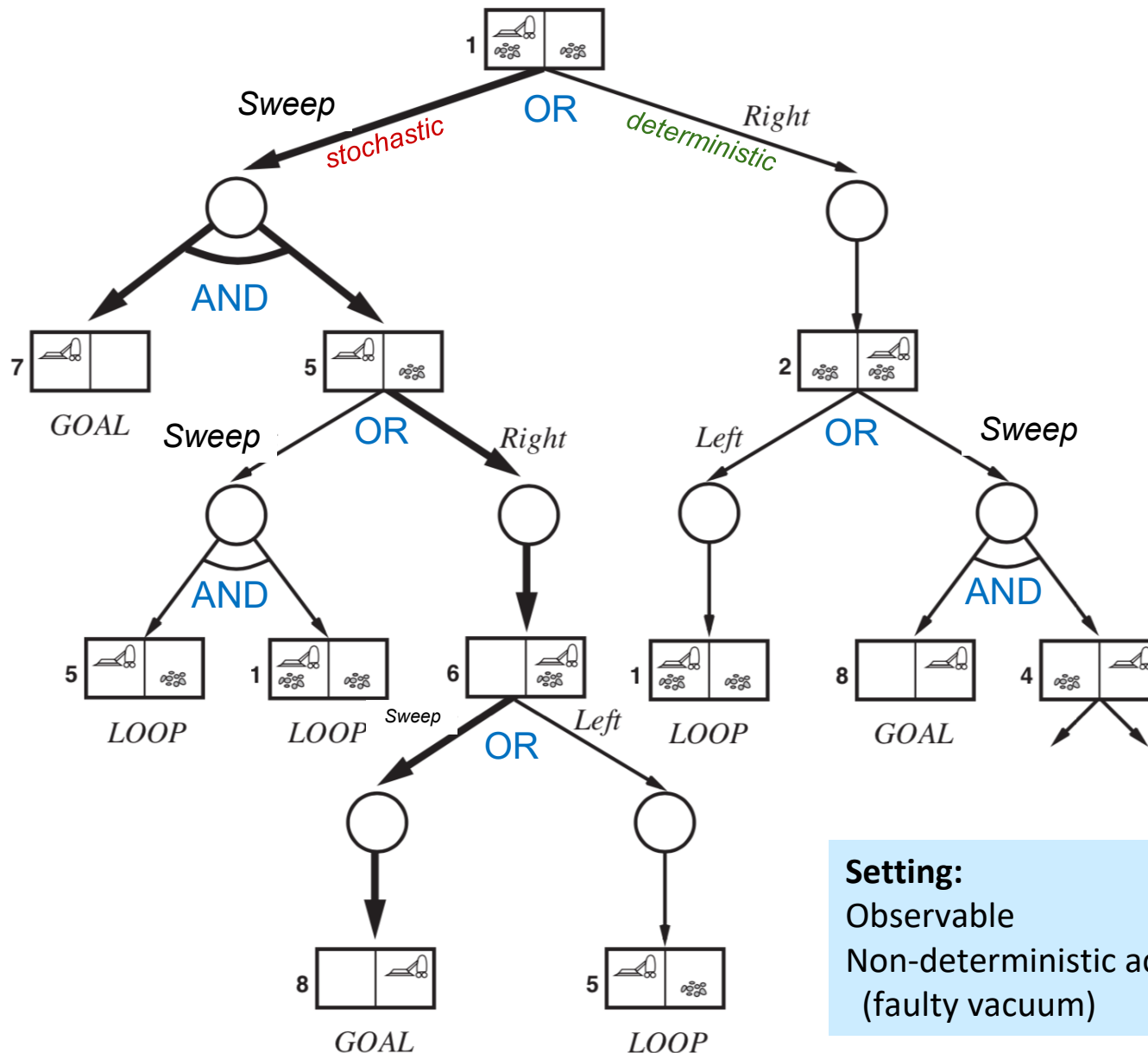
NB:
AND nodes only branch
 when the related action
 is stochastic.

Setting:
 Observable
 Non-deterministic actions
 (faulty vacuum)

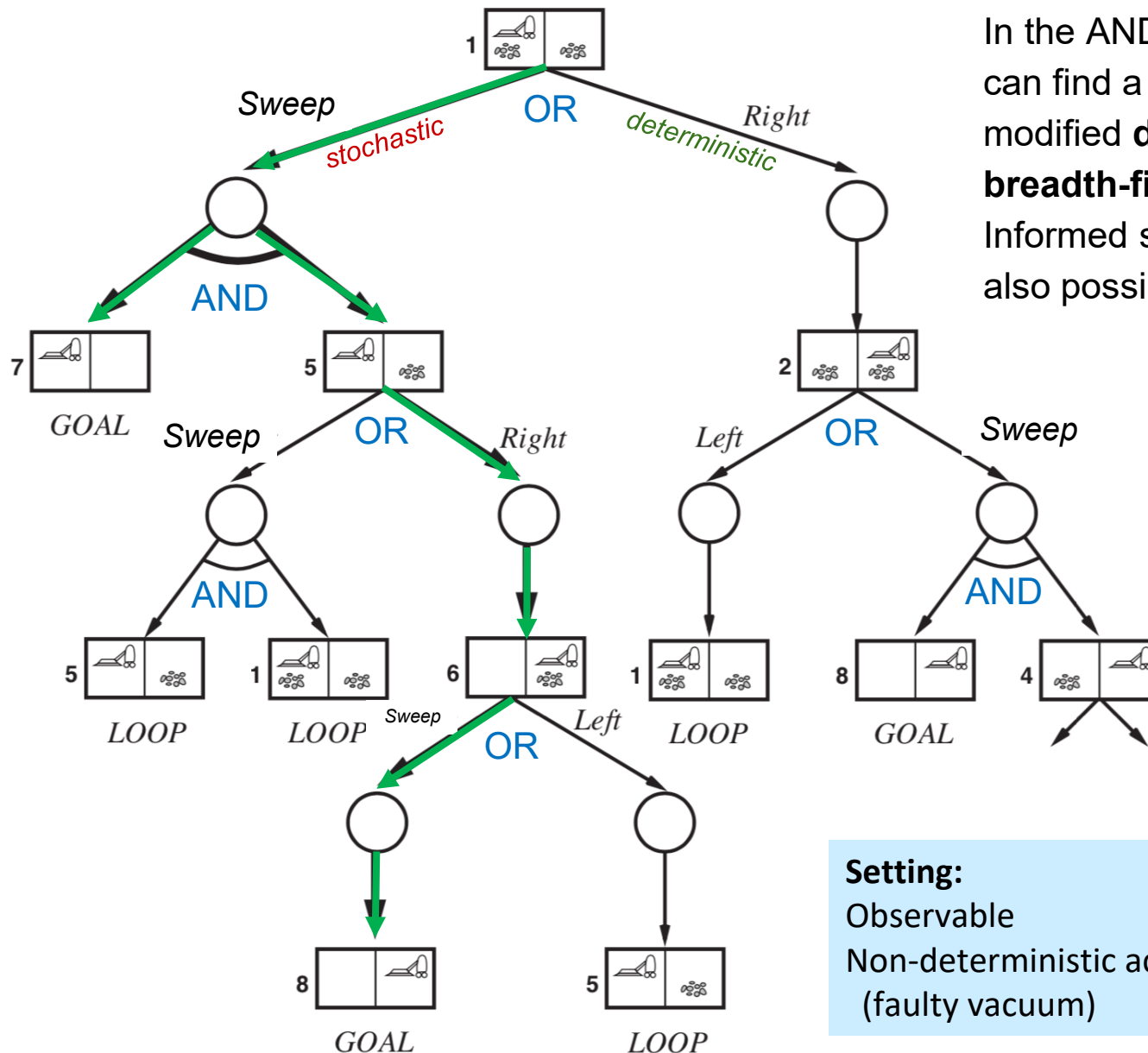
Non-deterministic vacuum: AND-OR search

A solution for an AND-OR search problem is a **subtree** that satisfies the following conditions:

1. It specifies **one action** at each **OR** node
2. It includes **every outcome** at each **AND** node
3. **It has a goal node at every leaf**



Setting:
 Observable
 Non-deterministic actions
 (faulty vacuum)



In the AND-OR setting we can find a solution with modified **depth-first**, or **breadth-first** strategy. Informed search strategies also possible.

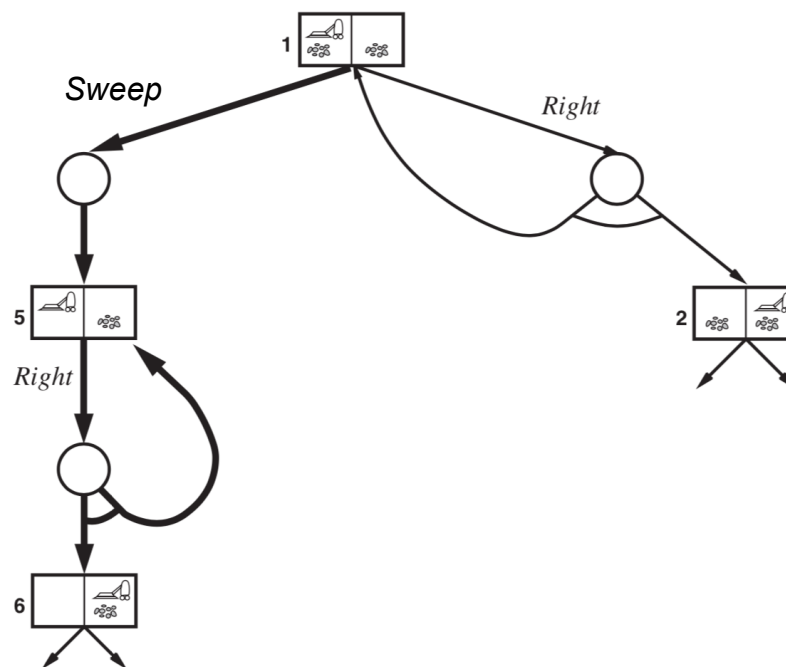
Setting:
Observable
Non-deterministic actions
(faulty vacuum)

Another case: The slippery vacuum

Movement actions, {Left, Right}, sometimes fail, leaving the agent in the same location. E.g. $\{1\} \xrightarrow{\text{Right}} \{1, 2\}$

Another case: The slippery vacuum

Movement actions, {Left, Right}, sometimes fail, leaving the agent in the same location. E.g., $\{1\} \xrightarrow{\text{Right}} \{1, 2\}$



There are no longer any acyclic solutions starting from State 1. Thus, AND-OR search will return with failure. What now?

Keep on trying until it works

Cyclic solution: keep on trying an action until it works.

Is this a good solution?

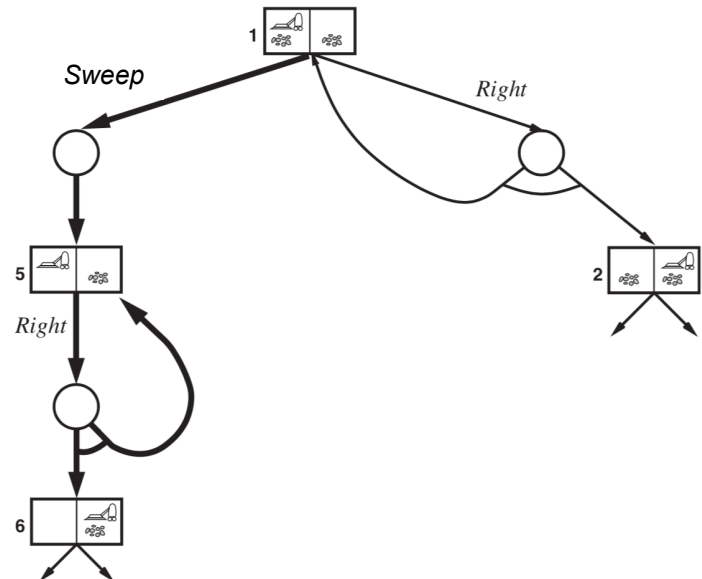
Depends on source of the stochasticity:

- OK if caused by a random event
 - e.g., Die roll
- Not OK if caused by an unobserved event
 - e.g., broken vacuum

Setting:

Observable

Non-deterministic actions
(slippery vacuum)

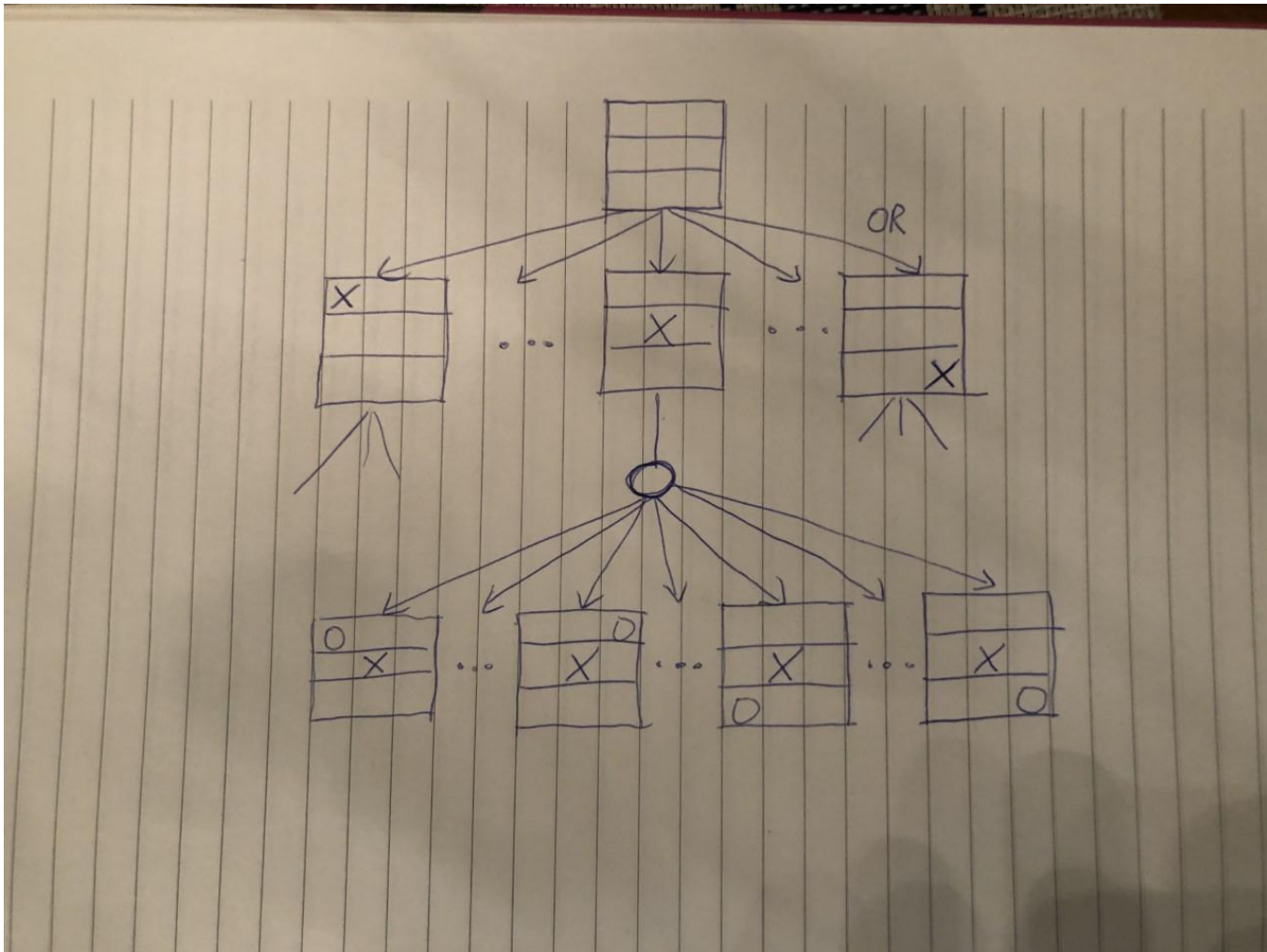


Exercise: Tic-Tac-Toe

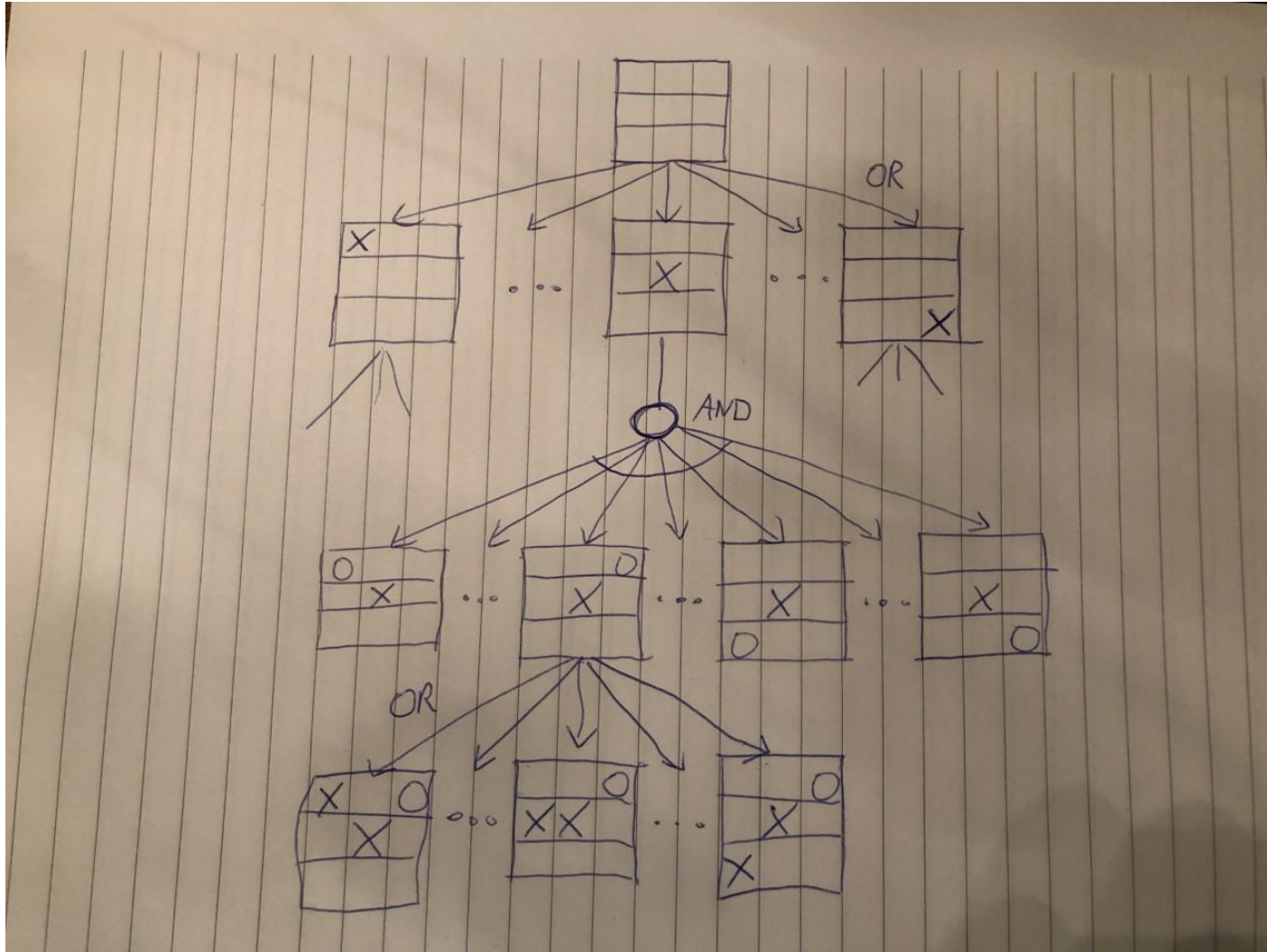
- What would the AND-OR tree for tic-tac-toe look like? Draw it for a few levels.
- What is a goal node?
- Suppose you are the starting player (X), and your opponent is (O).

Exercise: Tic-Tac-Toe

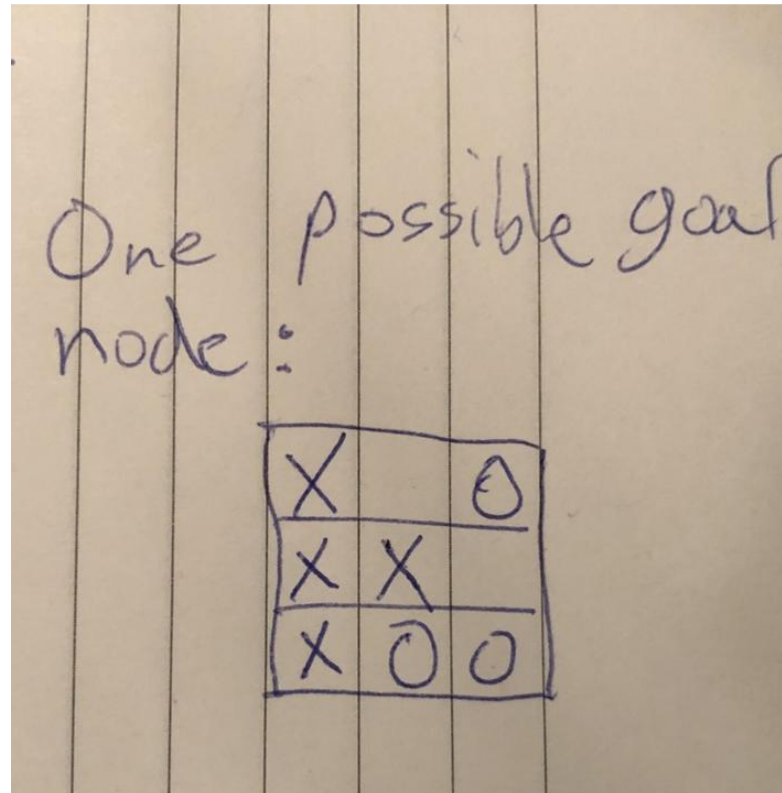
Exercise: Tic-Tac-Toe



Exercise: Tic-Tac-Toe



Exercise: Tic-Tac-Toe



Partial observability

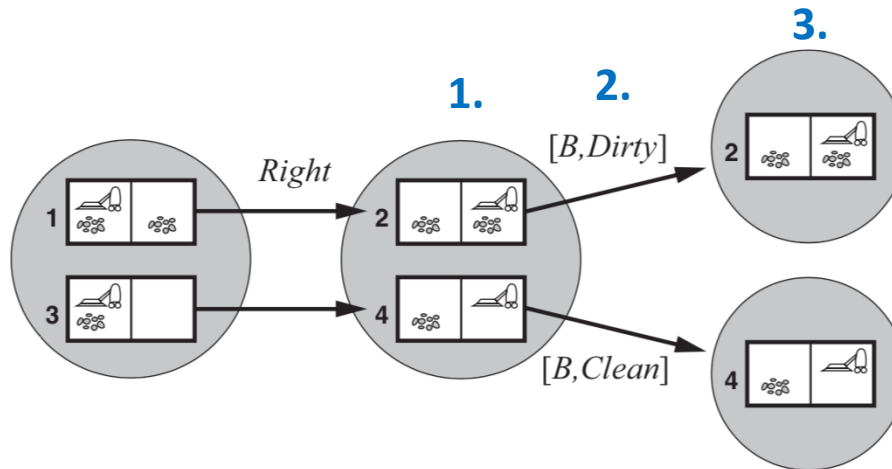
- **Partially observable** states lie between fully observable and unobservable states
- E.g., perhaps the vacuum can only sense locally: it senses if the current room is dirty or clean but **not** the adjacent room
- In this case, the agent must account for possible observations
- Observations (percepts) are very useful: they tell the agent something **partial** about its new state
- But they don't tell it everything. It must again maintain a **belief** over possibilities

Partial observability

- The problem formulation must specify how the environment generates observations from the state (*observation fcn*)
- Typically, several states can generate the same observation
- We can think of transitions from one belief state to the next for a given action as occurring in three stages:
 1. **prediction:** from the given action and the current belief, predict the next belief state (same as in unobservable case)
 2. **observation prediction:** predict which observations could be emitted in the predicted belief state
 3. **update:** for each possible observation, determine the resulting belief state; this is the set of states that could have produced the observation

Search with partial observations

- Deterministic case:

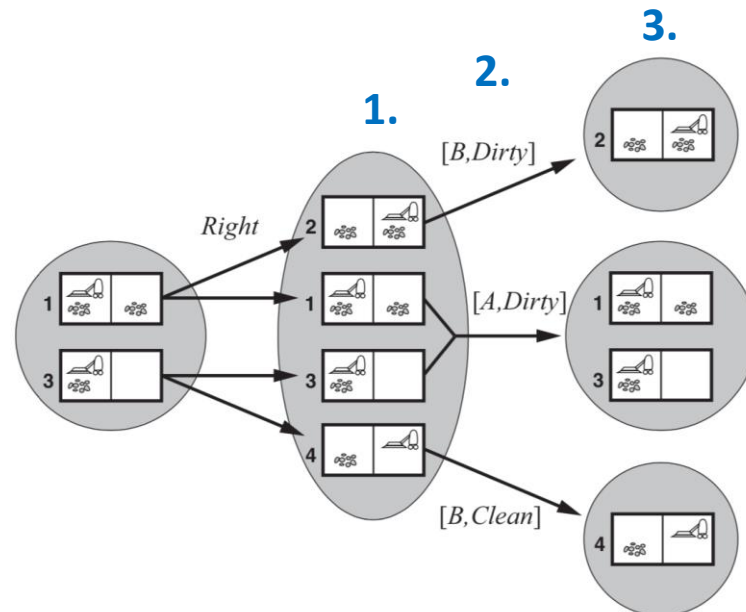


Setting:
Partially observable
(local sensing)
Deterministic actions

- Non-deterministic case:

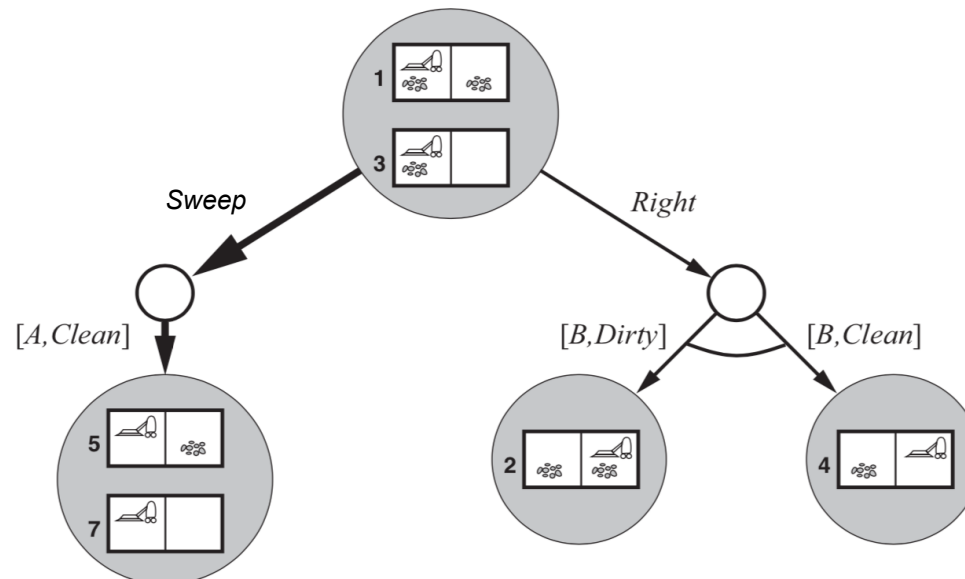
Setting:

Partially observable
(local sensing)
Non-deterministic actions
(slippery vacuum)



Searching the belief space

- In general, we can apply any standard search algorithm for partially observable problems
- E.g., AND-OR search for the local-sensing vacuum:



- AND-OR search returns a conditional plan that tests the **belief state** rather than the actual state

Searching the belief space

- In general, can apply any standard search algorithm for partially observable problems
- However, scalability is a major issue!
- **Problem #1:** Number of reachable beliefs can be very large
 - Use sampling or pruning to reduce this
- **Problem #2:** Number of physical states in each belief state can be very large
 - Use a compact state representation
 - Plan for each state separately (not as a search over belief space)

Questions

- What are the differences between the assumptions behind how we handle uncertainty during the last few lectures on game playing, vs. in this lecture?
- How do these differences propagate into the choice of representations, data structures, and algorithms?

Summary

- Actions can be {**deterministic**, **nondeterministic**}
- States can be {**fully observable**, **unobservable**, **partially observable (deterministic)**, **partially observable (non-deterministic)**}
- **Belief states** represent the set of possible physical states an agent is in
- AND-OR trees can be used to represent and search through a belief state space to produce a plan, even when there is uncertainty

Exercise: Mastermind

<http://www.web-games-online.com/mastermind/index.php>

- Belief space?
 - Initial belief?
 - Action space?
 - Deterministic / non-deterministic actions?
 - Possible percepts?
 - Perception function?
 - Goal test?
 - Step cost?
- 
- A wooden board game, possibly a version of the game of Go or a similar strategy game, is shown in the bottom right corner. The board is made of light-colored wood and features a grid of circular holes. Several colorful beads (yellow, orange, green, blue, purple, and white) are placed in the holes, arranged in a pattern that suggests a game in progress. The beads are small and round, and the board has a simple, clean design.

